EXPERIMENT 15:Experiment to understand the data preprocessing in Data science

DESCRIPTION:

The program demonstrates data preprocessing by handling missing values and converting categorical data into numerical form using a new dataset.

PROGRAM:

```
import pandas as pd

import numpy as np

# New dataset

data = {

    "City": ["Coimbatore", "Hyderabad", "Pune", "Hyderabad", "Kolkata",

            "Coimbatore", "Pune", np.nan, "Kolkata", "Coimbatore"],

    "Age": [22, 30, 26, 36, 42, np.nan, 33, 48, 29, 39],

    "Income": [28000, 55000, 38000, np.nan, 72000,

               32000, 49000, 80000, 45000, 65000],

    "Selected": ["Yes", "No", "Yes", "No", "Yes",

                 "No", "Yes", "No", "No", "Yes"]

}

df = pd.DataFrame(data)

print("Original Dataset:")

print(df)

# Handle missing values

df["City"] = df["City"].fillna(df["City"].mode()[0])

df["Age"] = df["Age"].fillna(df["Age"].median())

df["Income"] = df["Income"].fillna(round(df["Income"].mean()))

# Convert City into numerical columns

city = pd.get_dummies(df["City"])

# Create preprocessed dataset
```

```
updated_dataset = pd.concat(
    [city, df[["Age", "Income", "Selected"]]],
    axis=1
)
# Convert Selected into 0 and 1
updated_dataset["Selected"] = updated_dataset["Selected"].replace(
    {"No": 0, "Yes": 1}
)
print("\nPreprocessed Dataset:")
print(updated_dataset)
print("\nMissing Values:")
print(updated_dataset.isnull().sum())
```

OUTPUT:

```
Original Dataset:
         City   Age   Income Selected
0  Coimbatore  22.0  28000.0      Yes
1   Hyderabad  30.0  55000.0       No
2        Pune  26.0  38000.0      Yes
3   Hyderabad  36.0      NaN       No
4     Kolkata  42.0  72000.0      Yes
5  Coimbatore   NaN  32000.0       No
6        Pune  33.0  49000.0      Yes
7         NaN  48.0  80000.0       No
8     Kolkata  29.0  45000.0       No
9  Coimbatore  39.0  65000.0      Yes

Preprocessed Dataset:
   Coimbatore  Hyderabad  Kolkata   Pune   Age   Income Selected
0        True      False    False  False  22.0  28000.0        1
1       False       True    False  False  30.0  55000.0        0
2       False      False    False   True  26.0  38000.0        1
3       False       True    False  False  36.0  51556.0        0
4       False      False     True  False  42.0  72000.0        1
5        True      False    False  False  33.0  32000.0        0
6       False      False    False   True  33.0  49000.0        1
7        True      False    False  False  48.0  80000.0        0
8       False      False     True  False  29.0  45000.0        0
9        True      False    False  False  39.0  65000.0        1

Missing Values:
Coimbatore    0
Hyderabad     0
Kolkata       0
Pune          0
Age           0
Income        0
Selected      0
dtype: int64
```